

MaryNirmala Bandi

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Professional Summary

Data Analyst with 4+ years of experience and a Master's in Data Science, skilled in uncovering insights from complex datasets and delivering impactful visualizations. Proficient in SQL, Python, R, Power BI, Azure, and Qlik Sense, with experience across the full data lifecycle, cloud-based analytics, and data engineering collaboration. Knowledgeable and experienced working in Agile/DevOps environments.

Technical Skills

Computer: Microsoft Office(Excel, PowerPoint, Word), Visual Studio, GitHub, Web Development.

Visualization Tools: Power BI, Qlik Sense, Tableau, R Studio, ArcGis

Programming Languages: Python, R, DAX, HTML, CSS, JavaScript, jQuery

ML & AI Analytics: Orange Tool, MLLM Analysis, Supervised & Unsupervised Learning, Model Training, Natural Language Processing.

Cloud Platforms: Azure (Machine Learning, Event Hub, Data Lake, Storage, Functions, Synapse Analytics)

Databases: MySQL, MongoDB, Oracle

Experience

Tata Consultancy Services, Data Analyst, Developer.

Jan 2021 – Aug 2023

Hyderabad, Telangana, India.

- Built and maintained Qlik Sense dashboards with dynamic interactivity and access controls, reducing manual reporting time by 30%. Developed a custom web mashup page by integrating multiple apps into one by using HTML, CSS, JavaScript, and jQuery enhance user experience.
- Conducted performance testing across environments and collaborated with data engineering teams to optimize SQL/PLSQL queries. Performed data cleaning, transformation, and modeling with SQL, Power Query, and DAX, improving forecast accuracy by 20%.

University of Houston, Intern

May 2024 – June 2024

Houston, TX, USA

- Contributed to an energy transition research project using statistical modeling and machine learning in R Studio, Orange to enhance predictive insights. Built interactive dashboards using ArcGis, Power BI to visualize results and support policy analysis.

UHCL-Indian Student Association (ISA), Volunteer Web Developer

Dec 2023 – May 2025

Houston, TX, USA

- Built dynamic UI interactions using jQuery, AJAX, and modals to enhance responsiveness and user engagement. Improved form accuracy and user feedback by integrating SweetAlert notifications alongside robust JavaScript regex-based validation techniques.

Research Paper

Enhancing Adversarial Robustness in Image Classification Using MLLMs (Yet to be published)

Jan 2025 – May 2025

- Evaluated adversarial robustness of CNNs (ResNet50, DenseNet121), vision transformers (CLIP, BLIP), and multimodal LLMs (LLaVA) on the ImageNet dataset using advanced attacks (FGSM, BIM, PGD, MIM) and multi-stage pipelines (e.g., CLIP+LLaVA).
- Enhanced adversarial image classification accuracy to 95% by integrating NLP techniques (lemmatization, stemming) and Phi-2 LLM for semantic similarity scoring in caption-based reasoning.

Projects

Crime Trends Visualization for Chicago

Jul 2025

- Developed interactive Tableau dashboards to analyze crime trends and enforcement patterns in Chicago, enabling insights by time, crime type, arrest status, and highlighting high-risk periods and differences between violent and non-violent crimes.

Life Expectancy Analysis

Jan 2024 – May 2024

- Performed statistical analysis on global health data to determine key factors impacting life expectancy using regression, correlation, and PCA. Enhanced model accuracy and interpretability by reducing multicollinearity and dimensionality.

Analyzing Circadian Rhythms through Social Media Activity

Aug 2024 – Dec 2024

- Conducted activity analysis across various professional roles using timestamped social media data. Applied data pre-processing and visualization to identify the most active roles, time-based usage trends, and high-engagement locations.

Education

M.S in Data Science

May 2025

University of Houston-Clear Lake, Houston, TX, USA

GPA: 3.63

B.Tech in Electronics and Communication Engineering

Sep 2020

Vignan's Lara Institute of Technology and Science, Guntur, Andhra Pradesh, India

GPA: 7.50